Carpooling Coordination with Route Optimization

Team Details:

Student-1 : Piyush A Patel, SRN : PES2UG24CS350

Student-2 : Ritesh Paruchuri, SRN : PES2UG24CS340

DDL Query Screenshots:

```
piyush-a-patel@piyush:~/Windows/Piyush/Projects/Python/Carpooling Coordination with Route Optimization$ sudo -u postgres psql -f TEAM_03_DDL.sql
psql:TEAM_03_DDL.sql:1: NOTICE:  database "carpool" does not exist, skipping
DROP DATABASE
CREATE DATABASE
You are now connected to database "carpool" as user "postgres".
CREATE EXTENSION
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE INDEX
CREATE INDEX
CREATE INDEX
CREATE INDEX
CREATE INDEX
psql:TEAM_03_DDL.sql:91: NOTICE:  ivfflat index created with little data
DETAIL:  This will cause low recall.
HINT:  Drop the index until the table has more data.
CREATE INDEX
CREATE FUNCTION
CREATE TRIGGER
```

List of Tables:

```
piyush-a-patel@piyush:~/Windows/Piyush/Projects/Python/Carpooling Coordination with Route Optimization$ sudo -u postgres psql -d carpool
psql (16.15 (Ubuntu 16.15-1.pgdg24.04+2))
Type "help" for help.

carpool=# \dt
          List of relations
 Schema |   Name   | Type  |  Owner
--------+----------+-------+----------
 public | bookings | table | postgres
 public | ratings  | table | postgres
 public | rides    | table | postgres
 public | stops    | table | postgres
 public | users    | table | postgres
 public | vehicles | table | postgres
(6 rows)
```

Table 1 (USERS):

```
carpool=# \d users
                                        Table "public.users"
    Column     |            Type             | Collation | Nullable |                Default
---------------+-----------------------------+-----------+----------+---------------------------------------
 user_id       | integer                     |           | not null | nextval('users_user_id_seq'::regclass)
 name          | character varying(100)      |           | not null |
 email         | character varying(120)      |           | not null |
 phone         | character varying(15)       |           | not null |
 password_hash | character varying(255)      |           | not null |
 is_driver     | boolean                     |           |          | false
 created_at    | timestamp without time zone |           |          | CURRENT_TIMESTAMP
Indexes:
    "users_pkey" PRIMARY KEY, btree (user_id)
    "users_email_key" UNIQUE CONSTRAINT, btree (email)
    "users_phone_key" UNIQUE CONSTRAINT, btree (phone)
Referenced by:
    TABLE "bookings" CONSTRAINT "bookings_rider_id_fkey" FOREIGN KEY (rider_id) REFERENCES users(user_id) ON DELETE CASCADE
    TABLE "ratings" CONSTRAINT "ratings_rated_by_fkey" FOREIGN KEY (rated_by) REFERENCES users(user_id)
    TABLE "ratings" CONSTRAINT "ratings_rated_user_fkey" FOREIGN KEY (rated_user) REFERENCES users(user_id)
    TABLE "rides" CONSTRAINT "rides_driver_id_fkey" FOREIGN KEY (driver_id) REFERENCES users(user_id) ON DELETE CASCADE
    TABLE "vehicles" CONSTRAINT "vehicles_driver_id_fkey" FOREIGN KEY (driver_id) REFERENCES users(user_id) ON DELETE CASCADE
```

Table 2 (VEHICLES):

```
carpool=# \d VEHICLES
                                     Table "public.vehicles"
    Column    |         Type          | Collation | Nullable |                   Default
--------------+-----------------------+-----------+----------+----------------------------------------------
 vehicle_id   | integer               |           | not null | nextval('vehicles_vehicle_id_seq'::regclass)
 driver_id    | integer               |           | not null |
 make         | character varying(50) |           | not null |
 model        | character varying(50) |           | not null |
 plate_number | character varying(15) |           | not null |
 capacity     | smallint              |           | not null |
 color        | character varying(30) |           |          |
Indexes:
    "vehicles_pkey" PRIMARY KEY, btree (vehicle_id)
    "vehicles_plate_number_key" UNIQUE CONSTRAINT, btree (plate_number)
Check constraints:
    "vehicles_capacity_check" CHECK (capacity >= 1 AND capacity <= 8)
Foreign-key constraints:
    "vehicles_driver_id_fkey" FOREIGN KEY (driver_id) REFERENCES users(user_id) ON DELETE CASCADE
Referenced by:
    TABLE "rides" CONSTRAINT "rides_vehicle_id_fkey" FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id) ON DELETE CASCADE
```

Table 3 (RIDES):

```
carpool=# \d rides
                                        Table "public.rides"
     Column      |            Type             | Collation | Nullable |                Default
-----------------+-----------------------------+-----------+----------+---------------------------------------
 ride_id         | integer                     |           | not null | nextval('rides_ride_id_seq'::regclass)
 driver_id       | integer                     |           | not null |
 vehicle_id      | integer                     |           | not null |
 source          | character varying(150)      |           | not null |
 destination     | character varying(150)      |           | not null |
 departure_time  | timestamp without time zone |           | not null |
 available_seats | smallint                    |           | not null |
 price_per_seat  | numeric(8,2)                |           | not null |
 status          | character varying(20)       |           |          | 'scheduled'::character varying
 route_embedding | vector(16)                  |           |          |
 created_at      | timestamp without time zone |           |          | CURRENT_TIMESTAMP
Indexes:
    "rides_pkey" PRIMARY KEY, btree (ride_id)
    "idx_rides_driver" btree (driver_id)
    "idx_rides_route_embedding" ivfflat (route_embedding) WITH (lists='50')
    "idx_rides_status" btree (status)
Check constraints:
    "rides_available_seats_check" CHECK (available_seats >= 0)
    "rides_price_per_seat_check" CHECK (price_per_seat >= 0::numeric)
    "rides_status_check" CHECK (status::text = ANY (ARRAY['scheduled'::character varying, 'ongoing'::character varying, 'completed'::character varying, 'cancelled'::character varying]::text[
]))
Foreign-key constraints:
    "rides_driver_id_fkey" FOREIGN KEY (driver_id) REFERENCES users(user_id) ON DELETE CASCADE
    "rides_vehicle_id_fkey" FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id) ON DELETE CASCADE
Referenced by:
    TABLE "bookings" CONSTRAINT "bookings_ride_id_fkey" FOREIGN KEY (ride_id) REFERENCES rides(ride_id) ON DELETE CASCADE
    TABLE "stops" CONSTRAINT "stops_ride_id_fkey" FOREIGN KEY (ride_id) REFERENCES rides(ride_id) ON DELETE CASCADE
```

Table 4 (STOPS):

```
carpool=# \d stops
                                    Table "public.stops"
    Column     |          Type          | Collation | Nullable |                Default
---------------+------------------------+-----------+----------+---------------------------------------
 stop_id       | integer                |           | not null | nextval('stops_stop_id_seq'::regclass)
 ride_id       | integer                |           | not null |
 sequence_no   | smallint               |           | not null |
 location_name | character varying(150) |           | not null |
 latitude      | numeric(9,6)           |           | not null |
 longitude     | numeric(9,6)           |           | not null |
Indexes:
    "stops_pkey" PRIMARY KEY, btree (stop_id)
    "idx_stops_ride" btree (ride_id)
    "stops_ride_id_sequence_no_key" UNIQUE CONSTRAINT, btree (ride_id, sequence_no)
Foreign-key constraints:
    "stops_ride_id_fkey" FOREIGN KEY (ride_id) REFERENCES rides(ride_id) ON DELETE CASCADE
Referenced by:
    TABLE "bookings" CONSTRAINT "bookings_drop_stop_id_fkey" FOREIGN KEY (drop_stop_id) REFERENCES stops(stop_id)
    TABLE "bookings" CONSTRAINT "bookings_pickup_stop_id_fkey" FOREIGN KEY (pickup_stop_id) REFERENCES stops(stop_id)

carpool=#
```

Table 5 (BOOKINGS):

```
carpool=# \d bookings
                                        Table "public.bookings"
     Column     |            Type             | Collation | Nullable |                  Default
----------------+-----------------------------+-----------+----------+-------------------------------------------
 booking_id     | integer                     |           | not null | nextval('bookings_booking_id_seq'::regclass)
 ride_id        | integer                     |           | not null |
 rider_id       | integer                     |           | not null |
 pickup_stop_id | integer                     |           | not null |
 drop_stop_id   | integer                     |           | not null |
 seats_booked   | smallint                    |           | not null |
 booking_status | character varying(20)       |           |          | 'pending'::character varying
 booked_at      | timestamp without time zone |           |          | CURRENT_TIMESTAMP
Indexes:
    "bookings_pkey" PRIMARY KEY, btree (booking_id)
    "idx_bookings_ride" btree (ride_id)
    "idx_bookings_rider" btree (rider_id)
Check constraints:
    "bookings_booking_status_check" CHECK (booking_status::text = ANY (ARRAY['pending'::character varying, 'confirmed'::character varying, 'cancelled'::character varying, 'completed'::charac
ter varying]::text[]))
    "bookings_check" CHECK (pickup_stop_id <> drop_stop_id)
    "bookings_seats_booked_check" CHECK (seats_booked > 0)
Foreign-key constraints:
    "bookings_drop_stop_id_fkey" FOREIGN KEY (drop_stop_id) REFERENCES stops(stop_id)
    "bookings_pickup_stop_id_fkey" FOREIGN KEY (pickup_stop_id) REFERENCES stops(stop_id)
    "bookings_ride_id_fkey" FOREIGN KEY (ride_id) REFERENCES rides(ride_id) ON DELETE CASCADE
    "bookings_rider_id_fkey" FOREIGN KEY (rider_id) REFERENCES users(user_id) ON DELETE CASCADE
Referenced by:
    TABLE "ratings" CONSTRAINT "ratings_booking_id_fkey" FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE
Triggers:
    trg_booking_seats AFTER INSERT ON bookings FOR EACH ROW EXECUTE FUNCTION fn_update_seats()
```

Table 6 (RATINGS):

```
carpool=# \d ratings
                                      Table "public.ratings"
   Column   |            Type             | Collation | Nullable |                  Default
------------+-----------------------------+-----------+----------+-------------------------------------------
 rating_id  | integer                     |           | not null | nextval('ratings_rating_id_seq'::regclass)
 booking_id | integer                     |           | not null |
 rated_by   | integer                     |           | not null |
 rated_user | integer                     |           | not null |
 score      | smallint                    |           | not null |
 comment    | character varying(300)      |           |          |
 rated_at   | timestamp without time zone |           |          | CURRENT_TIMESTAMP
Indexes:
    "ratings_pkey" PRIMARY KEY, btree (rating_id)
    "ratings_booking_id_key" UNIQUE CONSTRAINT, btree (booking_id)
Check constraints:
    "ratings_check" CHECK (rated_by <> rated_user)
    "ratings_score_check" CHECK (score >= 1 AND score <= 5)
Foreign-key constraints:
    "ratings_booking_id_fkey" FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE
    "ratings_rated_by_fkey" FOREIGN KEY (rated_by) REFERENCES users(user_id)
    "ratings_rated_user_fkey" FOREIGN KEY (rated_user) REFERENCES users(user_id)
```